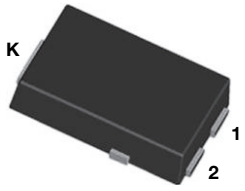
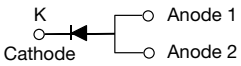
# Low Noise High Current Density Surface-Mount TMBS® (Trench MOS Barrier Schottky) Rectifier

Ultra Low $V_F$ = 0.49 V at $I_F$ = 5 A

**eSMP® Series**




**SMPC (TO-277A)**




## FEATURES

- Very low profile - typical height of 1.1 mm
- Trench MOS Schottky technology
- Low forward voltage drop, low power losses
- High efficiency operation
- Low reverse spike voltage
- Very low junction capacitance
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available
  - Automotive ordering code; base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE GRADE Available

RoHS COMPLIANT

HALOGEN FREE

## LINKS TO ADDITIONAL RESOURCES

3D Models | Packages | Marking

| PRIMARY CHARACTERISTICS | |
|---|---|
| $I_{F(AV)}$ | 10 A |
| $V_{RRM}$ | 100 V |
| $I_{FSM}$ | 180 A |
| $V_F$ at $I_F$ = 10 A (125 °C) | 0.57 V |
| $T_J$ max. | 175 °C |
| Package | SMPC (TO-277A) |
| Circuit configuration | Single |

## TYPICAL APPLICATIONS

For use in low voltage high frequency inverters, freewheeling, DC/DC converters, and polarity protection applications.

## MECHANICAL DATA

**Case:** SMPC (TO-277A)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-M3 - halogen-free, RoHS-compliant, and commercial grade
Base P/NHM3 - halogen-free, RoHS-compliant and AEC-Q101 qualified

**Terminals:** matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
M3 and HM3 suffix meet JESD 201 class 2 whisker test

| MAXIMUM RATINGS ($T_A$ = 25 °C unless otherwise noted) | | | |
|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **V10PM104** | **UNIT** |
| Device marking code | | 10M14 | |
| Maximum repetitive peak reverse voltage | $V_{RRM}$ | 100 | V |
| Maximum DC forward current | $I_{F(AV)}$ (1) | 10 | A |
| | $I_{F(AV)}$ (2) | 4 | |
| Peak forward surge current 10 ms single half sine-wave superimposed on rated load | $I_{FSM}$ | 180 | A |
| Operating junction temperature range | $T_J$ (3) | -40 to +175 | °C |
| Storage temperature range | $T_{STG}$ | -55 to +175 | °C |

**Notes**

(1) Mounted on 30 mm x 30 mm pad areas aluminum PCB
(2) Free air, mounted on recommended pad area
(3) The heat generated must be less than the thermal conductivity from junction to ambient: $dP_D/dT_J < 1/R_{\theta JA}$


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| ELECTRICAL CHARACTERISTICS ($T_J$ = 25 °C unless otherwise noted) | | | | | | |
|---|---|---|---|---|---|---|
| PARAMETER | TEST CONDITIONS | | SYMBOL | TYP. | MAX. | UNIT |
| Instantaneous forward voltage | $I_F$ = 5 A | $T_J$ = 25 °C | $V_F$ (1) | 0.56 | - | V |
| | $I_F$ = 10 A | | | 0.67 | 0.74 | |
| | $I_F$ = 5 A | $T_J$ = 125 °C | | 0.49 | - | |
| | $I_F$ = 10 A | | | 0.57 | 0.62 | |
| Reverse current | $V_R$ = 70 V | $T_J$ = 25 °C | $I_R$ (2) | 0.0004 | - | mA |
| | | $T_J$ = 125 °C | | 1 | - | |
| | $V_R$ = 100 V | $T_J$ = 25 °C | | - | 0.04 | |
| | | $T_J$ = 125 °C | | 2 | 6 | |
| Typical junction capacitance | 4.0 V, 1 MHz | | $C_J$ | 420 | - | pF |

**Notes**

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 5 ms

| THERMAL CHARACTERISTICS ($T_A$ = 25 °C unless otherwise specified) | | | |
|---|---|---|---|
| PARAMETER | SYMBOL | V10PM104 | UNIT |
| Typical thermal resistance | $R_{\theta JA}$ (1)(2) | 75 | °C/W |
| | $R_{\theta JM}$ (3) | 4 | |

**Notes**

(1) The heat generated must be less than the thermal conductivity from junction to ambient: $dP_D/dT_J < 1/R_{\theta JA}$

(2) Free air, mounted on recommended PCB, 2 oz. pad area; thermal resistance $R_{\theta JA}$ - junction to ambient

(3) Units mounted on 30 mm x 30 mm aluminum PCB, thermal resistance $R_{\theta JM}$ - junction to mount

## ORDERING INFORMATION TABLE

Device code

| V | 10 | P | M | 10 | 4 | H | M3 |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

1 - Vishay TMBS product

2 - Current rating (8 = 8 A)

3 - Package type (P = SMPC (TO-277A))

4 - Process type option (M = low $I_R$)

5 - Voltage rating (10 = 100 V)

6 - TMBS generation option (4 = gen 4)

7 - Quality grade (H = AEC-Q101 qualified, - = industry grade)

8 - Material / Environment category
(M3 = halogen-free, RoHS-compliant, and termination lead (Pb)-free)

| ORDERING INFORMATION (Example) | | | | |
|---|---|---|---|---|
| PREFERRED P/N | UNIT WEIGHT (g) | PACKAGE CODE | BASE QUANTITY | DELIVERY MODE |
| V10PM104-M3/H | 0.10 | H | 1500 | 7" diameter plastic tape and reel |
| V10PM104-M3/I | 0.10 | I | 6500 | 13" diameter plastic tape and reel |
| V10PM104HM3/H (1) | 0.10 | H | 1500 | 7" diameter plastic tape and reel |
| V10PM104HM3/I (1) | 0.10 | I | 6500 | 13" diameter plastic tape and reel |

**Note**

(1) AEC-Q101 qualified


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## RATINGS AND CHARACTERISTICS CURVES ($T_A$ = 25 °C unless otherwise specified)

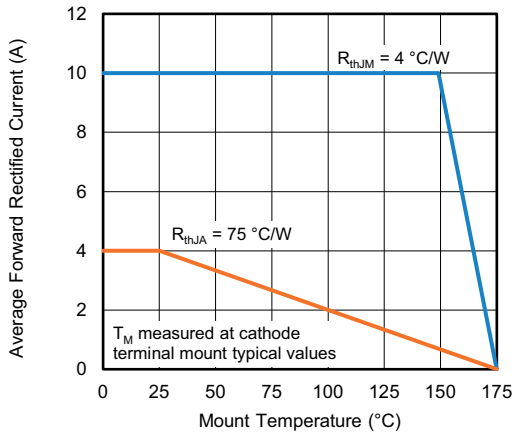



Fig. 1 - Maximum Forward Current Derating Curve

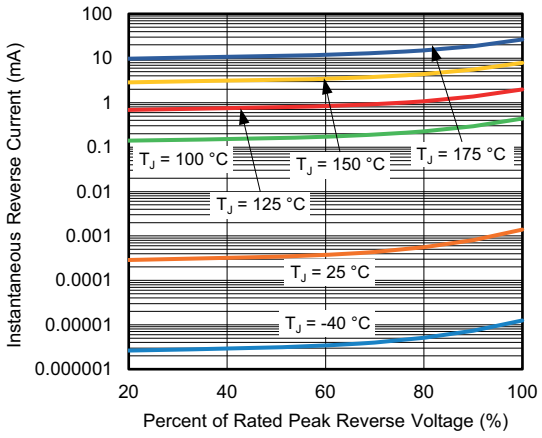



Fig. 4 - Typical Reverse Characteristics

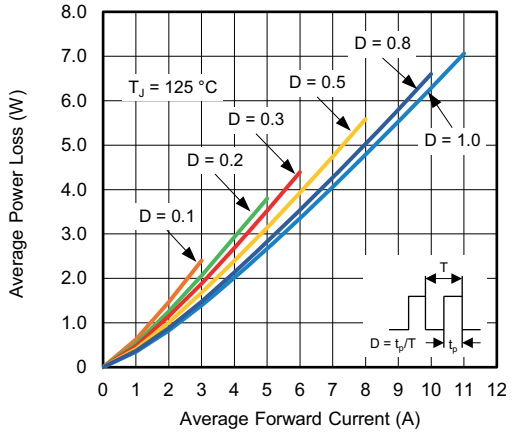



Fig. 2 - Forward Power Loss Characteristics

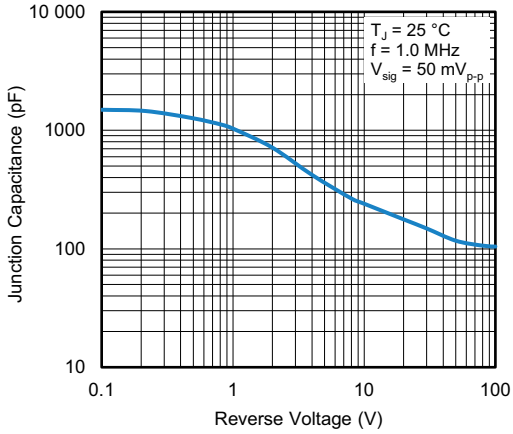



Fig. 5 - Typical Junction Capacitance

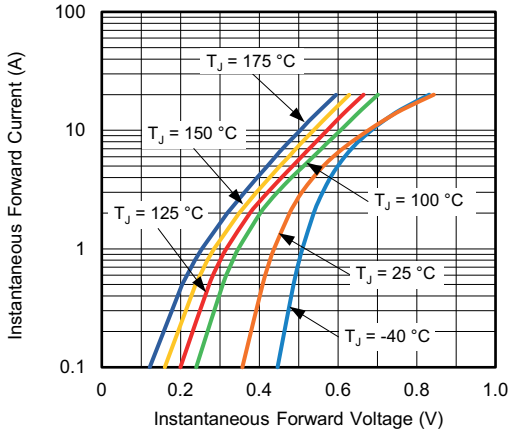



Fig. 3 - Typical Instantaneous Forward Characteristics

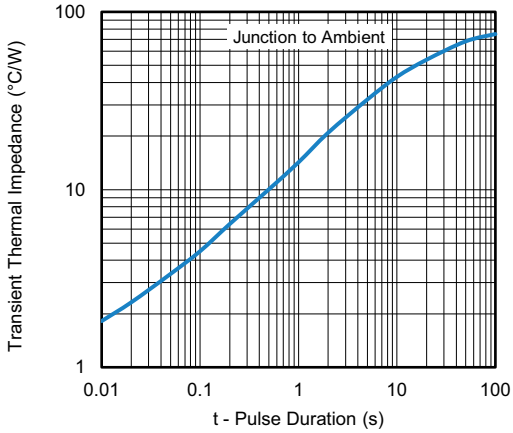
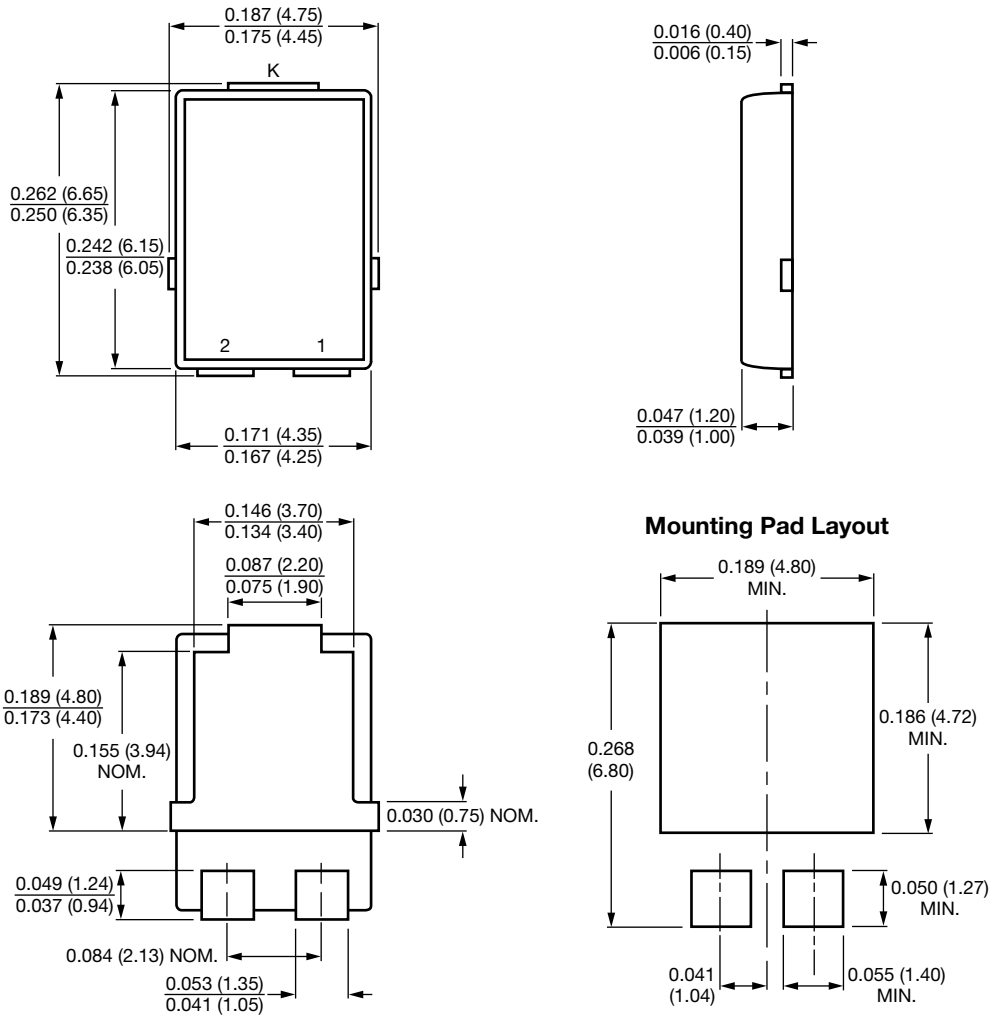



Fig. 6 - Typical Transient Thermal Impedance


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**PACKAGE OUTLINE DIMENSIONS** in inches (millimeters)

**SMPC (TO-277A)**




**Mounting Pad Layout**




Conform to JEDEC® TO-277A


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